

Jian Weng

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Education

University of California, Los Angeles

6th year Ph.D Candidate, Advisor: Tony Nowatzki

Los Angeles, CA

Aug. 2017 – Present

Shanghai Jiao Tong University

B.Eng. of Computer Science, ACM Honored Class

Shanghai, China

May 2013 – July 2017

Thesis: Compiling Domain Specific Language to Reconfigurable Accelerators

Conference Publications

- OverGen: Improving FPGA Usability through Domain-specific Overlay Generation.
Sihao Liu=, **Jian Weng**=, Dylan Kupsh, Atefeh Sohrabizadeh, Zhengrong Wang, Licheng Guo, Jiuyang Liu, Maxim Zhulin, Lucheng Zhang, Jason Cong, Tony Nowatzki.
International Symposium on Microarchitecture (MICRO), 2022, **Best Paper Runner-up**
- Near-Stream Computing: General and Transparent Near-Cache Acceleration.
Zhengrong Wang, **Jian Weng**, Sihao Liu, Tony Nowatzki.
International Symposium on High-Performance Computer Architecture (HPCA), 2022
- UNIT: Unifying Tensorized Instruction Compilation.
Jian Weng, Animesh Jain, Jie Wang, Leyuan Wang, Yida Wang, and Tony Nowatzki.
International Symposium on Code Generation and Optimization (CGO), 2021
- Stream Floating: Enabling Proactive and Decentralized Cache Optimizations.
Zhengrong Wang, **Jian Weng**, Jason Lowe-Power, Jayesh Gaur, Tony Nowatzki. *International Symposium on High-Performance Computer Architecture (HPCA)*, 2021, **Best Paper Runner-up**
- DSAGEN: Synthesizing Programmable Spatial Accelerators.
Jian Weng=, Sihao Liu=, Vidushi Dadu, Zhengrong Wang, Preyas Shah, and Tony Nowatzki.
International Symposium on Computer Architecture (ISCA), 2020, **IEEE Micro Honorable Mentions**
- A Hybrid Systolic-Dataflow Architecture for Inductive Matrix Algorithms.
Jian Weng, Sihao Liu, Zhengrong Wang, Vidushi Dadu, and Tony Nowatzki.
International Symposium on High-Performance Computer Architecture (HPCA), 2020
- Towards General Purpose Acceleration by Exploiting Common Data-Dependence Forms.
Vidushi Dadu, **Jian Weng**, Sihao Liu, and Tony Nowatzki.
International Symposium on Microarchitecture (MICRO), 2019 **IEEE Micro Top Picks**
- Hybrid Optimization/Heuristic Instruction Scheduling for Programmable Accelerator Codesign.
Tony Nowatzki, Newsha Ardalani, Karthikeyan Sankaralingam, and **Jian Weng**.
Parallel Architectures and Compilation Techniques (PACT), 2018

Journal Publications

- Unifying Spatial Accelerator Compilation with Idiomatic and Modular Transformations.
Jian Weng, Sihao Liu, Dylan Kupsh, Tony Nowatzki.
IEEE Micro Special Issue on Compiling for Accelerators, 2022
- DAEGEN: A Modular Compiler for Exploring Decoupled Access Execte Accelerators.
Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
Computer Architecture Letters (CAL), 2019

Workshops and Tutorials

- OverGen: Decoupled-Spatial Architecture Framework
Sihao Liu, **Jian Weng**, Dylan Kupsh, and Tony Nowatzki.
International Symposium on Microarchitecture (@MICRO'22)

- A Full-Stack Infrastructure for Automating Spatial Architecture Research
Jian Weng, Sihao Liu, Dylan Kupsh, and Tony Nowatzki.
Workshop on Democratizing Domain-Specific Accelerators (@MICRO'22)
- Generality is the Key Dimension in Accelerator Design
Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
Workshop on Languages, Tools, and Techniques for Accelerator Design (@ASPLOS'21)
- DSAGEN: Democratizing Decoupled Spatial Architecture Research
Jian Weng, Sihao Liu, Vidushi Dadu, and Tony Nowatzki.
International Symposium on Microarchitecture (@MICRO'20)

Invited Talks

Synthesizing Programmable Accelerators: A Full-Stack Perspective

- University of Illinois, Urbana Champaign, ILLIXR Consortium [↗](#) Nov. 9th, 2022
- Institute of Computing Technology, Chinese Academy of Sciences Nov. 19th, 2022
- Georgia Institute of Technology, EIC Lab Nov. 23rd, 2022

Synthesizing Programmable Accelerators: A Compiler Perspective

- University of Michigan, Computer Engineering Lab Feb. 23rd, 2022
- University of Illinois, Urbana Champaign, CAP Seminar [↗](#) Feb. 8th, 2022
- University of Washington, SAMPL Research Group [↗](#) Jan. 27th, 2022

Unit: Unifying Tensorized Instruction Compilation

- TVM Conference Dec. 2nd, 2020

Hybrid Script: A Text Format for Halide IR & A Python-TVM Hybrid Frontend

- TVM Conference Dec. 12th, 2018

Open-Source Projects and Infrastructures

Democratizing Spatial Accelerator Design [↗](#)

Co-first Author

Aug. 2017 – Now

- Develop the full-stack infrastructure for spatial architecture design.

TVM [↗](#)

Committer

Apache

Aug. 2017 – Now

- Implement the `tvm.te.hybrid` module to enhance the expressiveness of the framework.
- Write tutorials on performance tuning and customized compilation pass.

Professional Experiences

Amazon Web Service

Applied Scientist Intern, Mentor: Yida Wang

East Palo Alto

June 2019 – Sep. 2019, Jan. 2020 – Sep. 2020

- Compile the newly emerging tensorized instructions to diverse hardware platforms (CGO 2021).
- Automatically integrate cuDNN to a next-generation machine learning framework [↗](#).

Selected Awards

MICRO 2022 Best Paper Runner-up	Oct. 2022
UCLA Dissertation Year Fellowship	Sep. 2022
QualComm Fellowship Finalist	Aug. 2022
Facebook Fellowship Finalist	April 2022
UCLA Graduate Student Fellowship	Nov. 2017, Mar. 2021
IEEE Micro Top Picks Honorable Mention	2021
HPCA Best Paper Runner-up	2021
IEEE Micro Top Picks	2020

Teaching

CS33: Introduction to Computer Organization

Teaching Assistant, w/ Prof. Glenn Reinman

MS109: Compiler Design and Implementation

Teaching Assistant, w/ Prof. Yong Yu

MS105: Data Structure

Teaching Assistant, w/ Prof. Huiyu Weng

University of California, Los Angeles

Mar. 24th — Jun. 11th, 2021

Shanghai Jiao Tong University

Mar. 22nd — Jun. 26th, 2016

Shanghai Jiao Tong University

Sep. 14th, 2015 — Jan. 17th, 2016

Research Mentoring

Rishabh Mani

Develop memory access pattern analysis; coauthor of OverGen.

Dylan Kupsh

Study spatial architecture mapping techniques; coauthor of OverGen.

Siyuan Ma

Study in-memory processing; submitted to ISCA'23.

Shuo Wang

Study concurrent application mapping to DSAGEN.

University of California, Los Angeles

Mar. 23rd, 2022 — Now

University of California, Los Angeles

Sep. 20th, 2021 — Now

University of Texas, Austin

Mar. 23rd, 2022 — Now

University of California, Los Angeles

Mar. 23rd, 2022 — Now